

DPT Exercises

Exercise sheet 1

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Exercise 1. True or false? Justify your answers either by a proof or a counterexample.

- (i) For a probability space $(\Omega, \mathcal{A}, P)$ we have $P(\Omega \setminus A) = 1 - P(A)$ for all $A \in \mathcal{A}$.
- (ii) Let $(\Omega, \mathcal{A}, \mu)$ be a measure space, $A, B \in \mathcal{A}$ such that $\mu(A \cap B) < \infty$. Then $\mu(B \setminus A) = \mu(B) - \mu(A \cap B)$.
- (iii) For a probability space $(\Omega, \mathcal{A}, P)$ and a sequence $(A_i)_{i \in \mathbb{N}}$ in $\mathcal{A}$, we have

$$P\left(\bigcup_{i \in \mathbb{N}} A_i\right) = \sup_{i \in \mathbb{N}} P(A_i).$$

Let $S \subseteq \mathbb{R}$. A real number $u \in \mathbb{R}$ is called an *upper bound* of S if $x \leq u$ for all $x \in S$. An upper bound s of S is called *supremum* or *least upper bound* if for all upper bounds u of S in $\mathbb{R}$ we have $s \leq u$. We then write $\sup(S) = s$. When $S = \{x_i \mid i \in \mathbb{N}\} \subseteq \mathbb{R}$ is at most countable, we also write $\sup_{i \in \mathbb{N}} x_i = s$. **Suprema and infima** can be defined for subsets of any partially ordered set.

- (iv) Let $(\Omega, \mathcal{A}, P)$ be a probability space and $A, B, C \in \mathcal{A}$. Then

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

- (v) The triple consisting of $\mathbb{N}$, $\mathcal{P}(\mathbb{N})$, and the set function

$$\mu : \mathcal{P}(\mathbb{N}) \rightarrow [0, \infty], A \mapsto \begin{cases} 0 & A \text{ finite} \\ \infty & A \text{ infinite} \end{cases}$$

form a measure space.

- (vi) We consider the two-time fair coin flip, where we interpret the two sides to count as 0 or 1. We are only interested in the sum of the two outcomes. Then the cdf is given by

$$F(x) = \begin{cases} 0 & x < 0, \\ \frac{1}{4} & x \in [0, 1), \\ \frac{3}{4} & x \in [1, 2), \\ 1 & x \geq 2. \end{cases}$$

Solution:

- (i) **True.** This is, because P is σ -additive (thus also finitely additive) and since A and $\Omega \setminus A$ are disjoint we have $1 = P(\Omega) = P(A \cup (\Omega \setminus A)) = P(A) + P(\Omega \setminus A)$.
- (ii) **True.** As $B = (B \setminus A) \dot{\cup} (A \cap B)$ the additivity of μ gives

$$\mu(B) = \mu(B \setminus A) + \mu(A \cap B)$$

and by subtracting $\mu(A \cap B)$, we obtain the statement.

- (iii) **False.** A possible counter-example would be: Consider the probability space $(\{0, 1\}, \mathcal{P}(\{0, 1\}), P)$ with $P(\{0\}) = P(\{1\}) = \frac{1}{2}$. For $A_0 = \{0\}$ and $A_n = \{1\}$ for each $n > 0$ we have

$$P\left(\bigcup_{n \in \mathbb{N}} A_n\right) = P(\{0, 1\}) = 1 \neq \frac{1}{2} = \sup_{n \in \mathbb{N}} P(A_n) = \sup_{n \in \mathbb{N}} \left(\frac{1}{2}\right).$$

(iv) **True.** It is a special case of the below inclusion-exclusion formula.

(v) **False.** This can be seen by choosing $A_n = \{n\}$, $n \in \mathbb{N}$ and noting that

$$\infty = \mu(\mathbb{N}) = \mu\left(\bigcup_{n=0}^{\infty} A_n\right) \neq \sum_{n=0}^{\infty} \mu(A_n) = 0$$

therefore not satisfying the σ -additivity.

(vi) **True.** We can model the experiment with $\Omega = \{0, 1, 2\}$ with probability measure that satisfies $P(\{0\}) = P(\{2\}) = \frac{1}{4}$, $P(\{1\}) = \frac{1}{2}$ and obtain

$$F(x) := P((-\infty, x]) = \sum_{n \in \{0, 1, 2\}, n \leq x} P(\{n\})$$

Exercise 2. Let $\Omega = \{\square, \square, \square, \square, \square, \square\}$, representing the rolling of a fair die. Assume a uniform distribution over Ω and let $p \in [0, 1]$ be the probability of obtaining at least one $\square$ when throwing six fair dice at once. Compute p (using rigorous mathematical notation, i.e., setting up the probability spaces properly).

Solution: The probability space for Ω^6 equipped with the uniform distribution is given by $(\{\square, \dots, \square\}^6, \mathcal{P}(\{\square, \dots, \square\}^6), P)$ for

$$P : \mathcal{P}(\{\square, \dots, \square\}^6) \rightarrow [0, 1], \quad A \mapsto \frac{|A|}{6^6}.$$

The event we care about is

$$A = \{\omega \in \{\square, \dots, \square\}^6 \mid \omega_j = \square \text{ for at least one } j \in \{1, \dots, 6\}\}.$$

Now we compute

$$\begin{aligned} p &= P(A) \\ &= 1 - P(\{\square, \dots, \square\}^6 \setminus A) \\ &= 1 - P(\{\omega \in \{\square, \dots, \square\}^6 \mid \omega_j \neq \square \text{ for all } j \in \{1, \dots, 6\}\}) \\ &= 1 - \frac{|\{\square, \square, \square, \square, \square, \square\}|}{|\{\square, \dots, \square\}^6|} \\ &\approx 0.665. \end{aligned}$$

Exercise 3 (Inclusion-exclusion formula). Let $(\Omega, \mathcal{A}, P)$ be a probability space, $(A_i)_{i \in [n]} \in \mathcal{A}^{[n]}$ for a given $n \in \mathbb{N}$. Then

$$P(A_1 \cup \dots \cup A_n) = \sum_{k=1}^n (-1)^{k-1} \sum_{1 \leq i_1 < \dots < i_k \leq n} P(A_{i_1} \cap \dots \cap A_{i_k})$$

In particular, it holds that

$$P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2).$$

[Hint: Use complete induction.]

Bonus: Show that moreover

$$P(A_1 \cap \dots \cap A_n) = \sum_{k=1}^n (-1)^{k-1} \sum_{\{i_1, \dots, i_k\} \subseteq [n]} P(A_{i_1} \cup \dots \cup A_{i_k}).$$

Solution: The proof works by complete induction using the difference rule.

(IH): $n = 2$: As $A_1 \cup A_2 = (A_1 \setminus A_2) \dot{\cup} (A_2)$, additivity and the difference rule yield:

$$P(A_1 \cup A_2) = P(A_2) + P(A_1 \setminus A_2) = P(A_2) + P(A_1) - P(A_1 \cap A_2).$$

(IH): The statement holds for n .

(IS): $n \rightarrow n+1$: As $A_1 \cup \dots \cup A_n \cup A_{n+1} = (A_1 \cup \dots \cup A_n) \dot{\cup} (A_{n+1} \setminus (A_1 \cup \dots \cup A_n))$, we obtain:

$$P(A_1 \cup \dots \cup A_n \cup A_{n+1}) = P(A_1 \cup \dots \cup A_n) + P(A_{n+1} \setminus (A_1 \cup \dots \cup A_n)).$$

Applying (IH) to $\bigcup_{i=1}^n A_i$ and $\bigcup_{i=1}^n (A_i \cap A_{n+1})$ and a slight change of notation yields:

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{k=1}^n (-1)^{k-1} \sum_{i_1, \dots, i_k \in [n]} P(A_{i_1} \cap \dots \cap A_{i_k})$$

as well as

$$\begin{aligned} P\left(\bigcup_{i=1}^n (A_i \cap A_{n+1})\right) &= \sum_{k=1}^n (-1)^{k-1} \sum_{\{i_1, \dots, i_k\} \subseteq [n]} P(A_{i_1} \cap \dots \cap A_{i_k} \cap A_{n+1}) \\ &= \sum_{k=1}^n (-1)^{k-1} \sum_{\substack{\{i_1, \dots, i_{k+1}\} \subseteq [n+1], \\ n+1 \in \{i_1, \dots, i_{k+1}\}}} P(A_{i_1} \cap \dots \cap A_{i_k} \cap A_{i_{k+1}}). \end{aligned}$$

Note that in the first expression, for fixed k , the sum goes over all sets $\{i_1, \dots, i_k\} \subseteq [n]$. On the other hand, in the second expression, the sum for k goes over all sets $\{i_1, \dots, i_{k+1}\} \subseteq [n+1]$ such that $n+1 \in \{i_1, \dots, i_{k+1}\}$. Thus, adding both sums for $k = 1, \dots, n$ gives:

$$\sum_{\{i_1, \dots, i_k\} \subseteq [n]} P(A_{i_1} \cap \dots \cap A_{i_k}) + \sum_{\substack{\{i_1, \dots, i_k\} \subseteq [n+1] \\ n+1 \in \{i_1, \dots, i_k\}}} P(A_{i_1} \cap \dots \cap A_{i_k}) = \sum_{\{i_1, \dots, i_k\} \subseteq [n+1]} P(A_{i_1} \cap \dots \cap A_{i_k}).$$

Plugging these results into the expression yields:

$$\begin{aligned} P(A_1 \cup \dots \cup A_n \cup A_{n+1}) &= P(A_{n+1}) + \sum_{k=1}^n (-1)^{k-1} \sum_{i_1, \dots, i_k \subseteq [n]} P(A_{i_1} \cap \dots \cap A_{i_k}) \\ &\quad + \sum_{k=1}^n (-1)^k \sum_{\substack{\{i_1, \dots, i_{k+1}\} \subseteq [n+1] \\ n+1 \in \{i_1, \dots, i_{k+1}\}}} P(A_{i_1} \cap \dots \cap A_{i_k} \cap A_{i_{k+1}}) \end{aligned}$$

and shifting the index $k \rightarrow k - 1$ in the second summation gives:

$$\begin{aligned} P(A_{n+1}) + \sum_{k=1}^n (-1)^{k-1} \sum_{i_1, \dots, i_k \subseteq [n]} P(A_{i_1} \cap \dots \cap A_{i_k}) + \sum_{k=2}^{n+1} (-1)^{k-1} \sum_{\substack{\{i_1, \dots, i_k\} \subseteq [n+1] \\ n+1 \in \{i_1, \dots, i_k\}}} P(A_{i_1} \cap \dots \cap A_{i_k}) \\ = \sum_{k=1}^{n+1} (-1)^{k-1} \sum_{\{i_1, \dots, i_k\} \subseteq [n+1]} P(A_{i_1} \cap \dots \cap A_{i_k}). \end{aligned}$$

Exercise 4. The DWT tutor Angelika supervises an exercise group with n sheets being submitted anonymously. After grading, Angelika randomly redistributes the sheets. What is the probability that no one gets their original sheet back? How does this probability behave as $n \rightarrow \infty$?

[Hint: Use the inclusion-exclusion formula and that $\exp(x) = \sum_{k=0}^{\infty} \frac{x^k}{k!}$]

Solution: The experiment is equivalent to drawing n distinct balls without replacement in order from an urn (Lemma 1.33 “without replacement ordered draw”). Thus, we have:

$$\Omega = \{\omega = (\omega_1, \dots, \omega_n) \in [n]^n \mid \omega_i \neq \omega_j \forall i \neq j\}, \quad |\Omega| = n!.$$

The probability of the event “the i -th candidate gets its original sheet,” $A_i = \{\omega \in [n]^n \mid \omega_i = i\}$, is

$$P(A_i) = \frac{|A_i|}{|\Omega|} = \frac{(n-1)!}{n!} = \frac{1}{n}.$$

Moreover, for $A_{i_1}, \dots, A_{i_k}$ where the i_j ’s are all distinct, we have:

$$P(A_{i_1} \cap \dots \cap A_{i_k}) = \frac{1}{n!} \cdot (n-k)!.$$

Then, by the inclusion-exclusion formula

$$P(A_1 \cup \dots \cup A_n) = \sum_{k=1}^n (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} P(A_{i_1} \cap \dots \cap A_{i_k}),$$

and as we have $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ possibilities to draw k out of n sheets, we obtain:

$$P(A_1 \cup \dots \cup A_n) = \sum_{k=1}^n \frac{(-1)^{k+1}}{k!}.$$

Thus, the probability that no one gets their original sheet back is:

$$1 - P(A_1 \cup \dots \cup A_n) = 1 - \sum_{k=1}^n \frac{(-1)^{k+1}}{k!} = \sum_{k=0}^n \frac{(-1)^k}{k!}.$$

In the limit, using the exponential series $e^{-1} = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!}$, we obtain:

$$\lim_{n \rightarrow \infty} \left(1 - P \left(\bigcup_{i=1}^n A_i \right) \right) = \frac{1}{e}.$$

